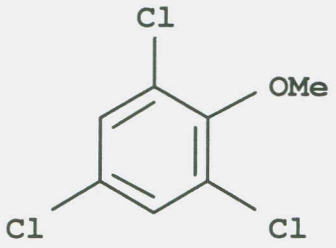
February 20, 2009 Analytical Chemistry Cumulative Exam.

This question is adapted from *J. Chromatography A*, Vol 1107, pp 240-247.

" Ever since Dom Perignon, a French Benedictine monk, began to use bark oak (*Quercus suber*) to close wine bottles, the cork-wine combination has prevailed, because the cork stopper has been regarded as the most suitable material to preserve wine because of its physical properties and quality aspects. However, cork is a natural product and not inert, so it can interact with wine, sometimes modify its flavour and even give it organoleptic defects."

The principal chemical responsible for this off flavor and smell is 2,4,6-trichloroanisole (TCA) which is illustrated below. Typically, TCA makes up about 80% of the chemical compounds that lead to the off flavor and smell. Other chemicals include 2,3,4,6-tetrachloroanisole (TeCA) and 2,3,4,5,6-pentachloroanisole (PCA). These compounds arise due to a variety of disinfectant practices. These compounds that are present in cork migrate into the wine leaving bad odor and off flavors. The sensorial perception (e.g., smell) of these compounds is 1 ng/L to 50 ng/L



a) Starting with just the cork, describe a sampling method or methods for collecting the chloroanisoles that will be compatible with the overall analytical method you outline in "c". Justify your reasoning.

b) Internal standards are common for many methods. What would be the purpose of an internal standard for this particular analysis? Draw or name a potential internal standard for this application. Be sure to justify your reasoning.

c) What would be an appropriate instrumental method for quantitation of these analytes? Be sure to justify the use and advantages of your proposed method over other potential methods for solving this chemical analysis problem.

d) Briefly describe how you would address method performance characteristics including calibration, analyte recovery from the cork, LOD, LOQ, and method repeatability.